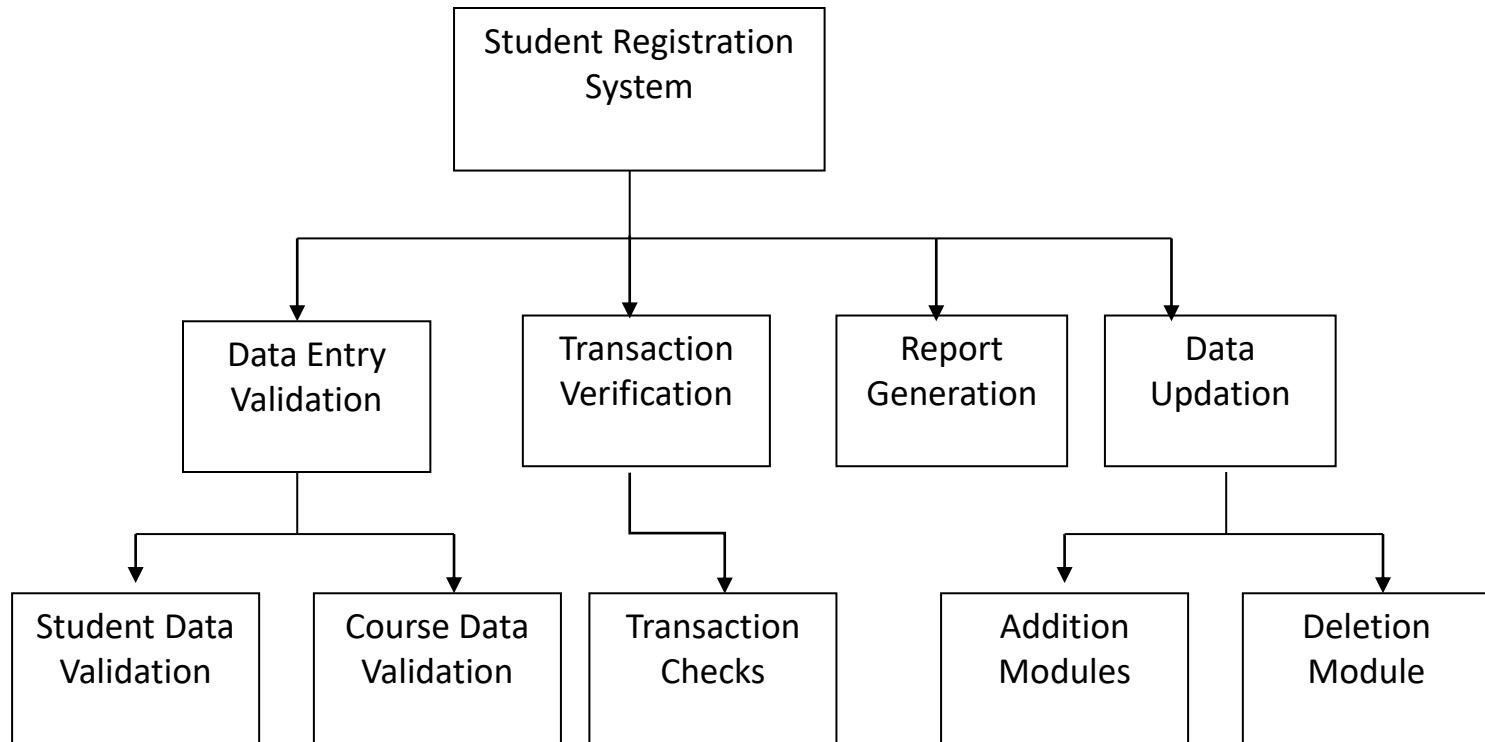


Component based Design

Zeeshan Rana

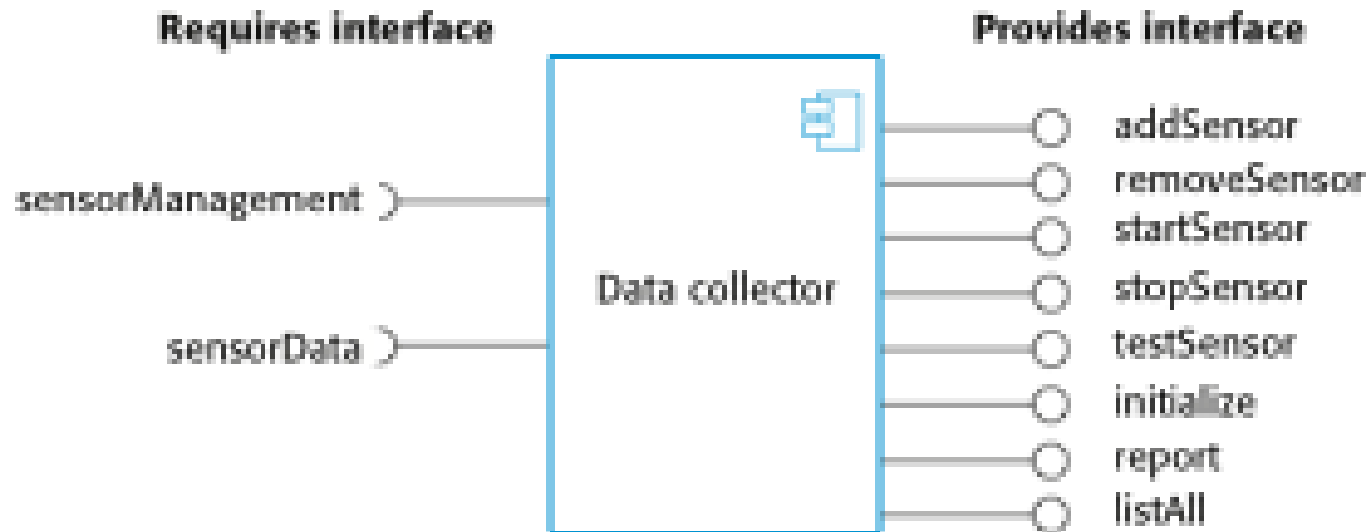
Subsystem Decomposition



Logical component: business components, logic layer, no explicit physical run-time equivalent

Physical component: database server, explicit run-time equivalent

A Component with Interfaces



Component Diagram

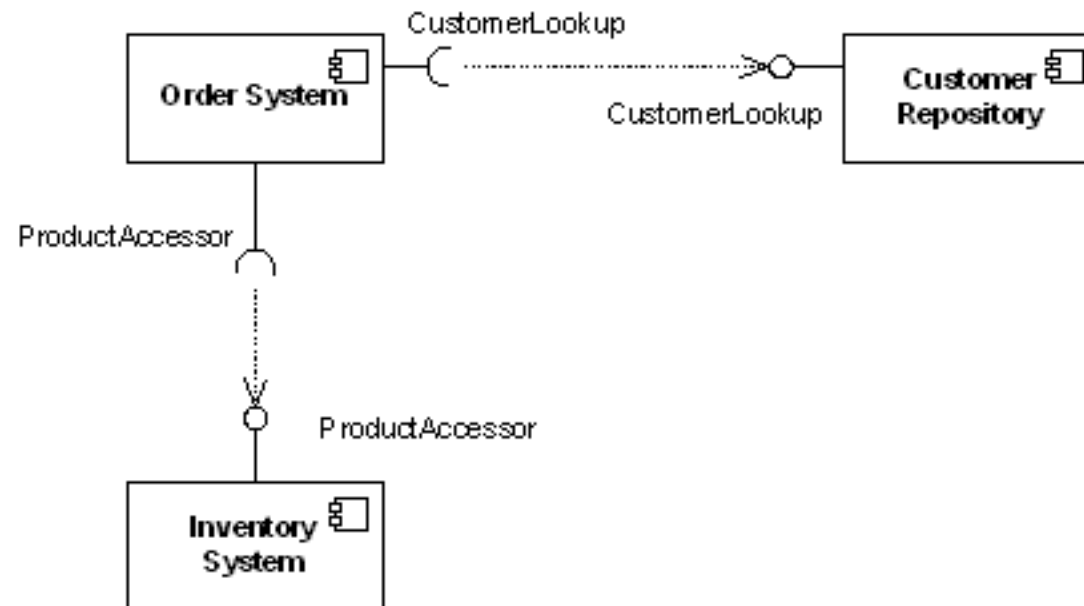
UML1.x



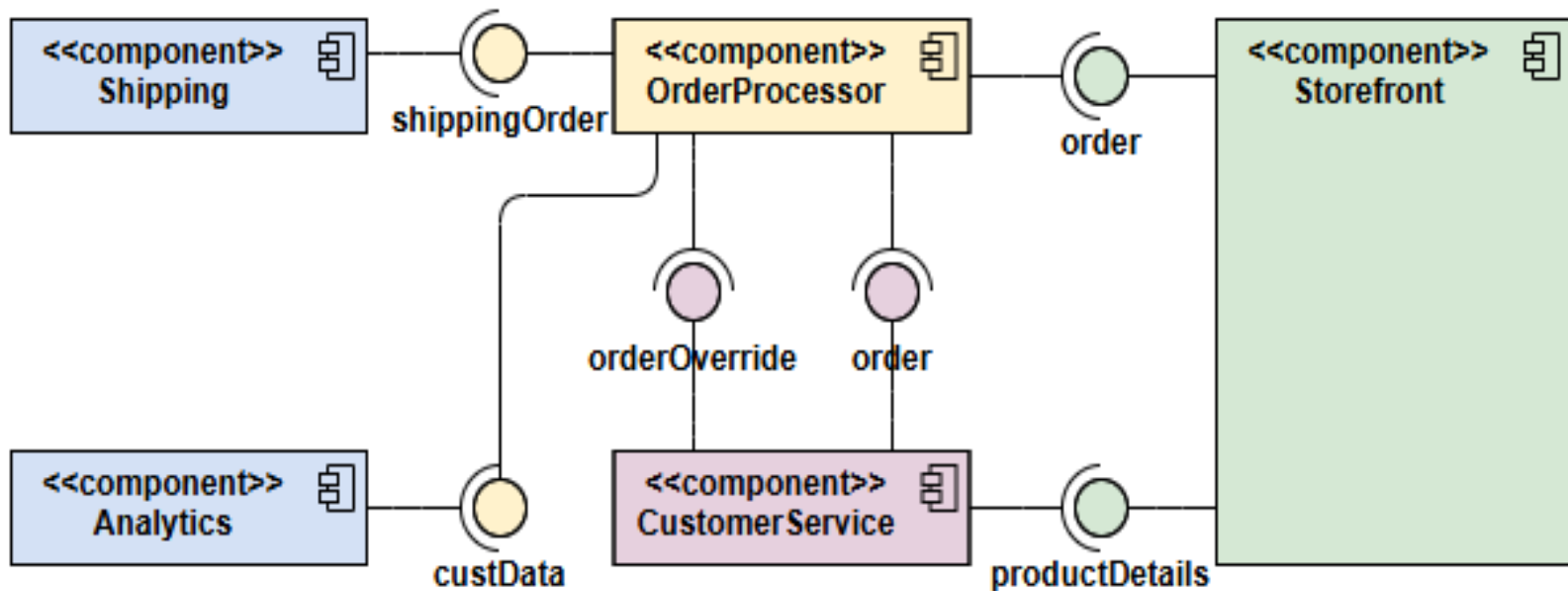
UML 2.0



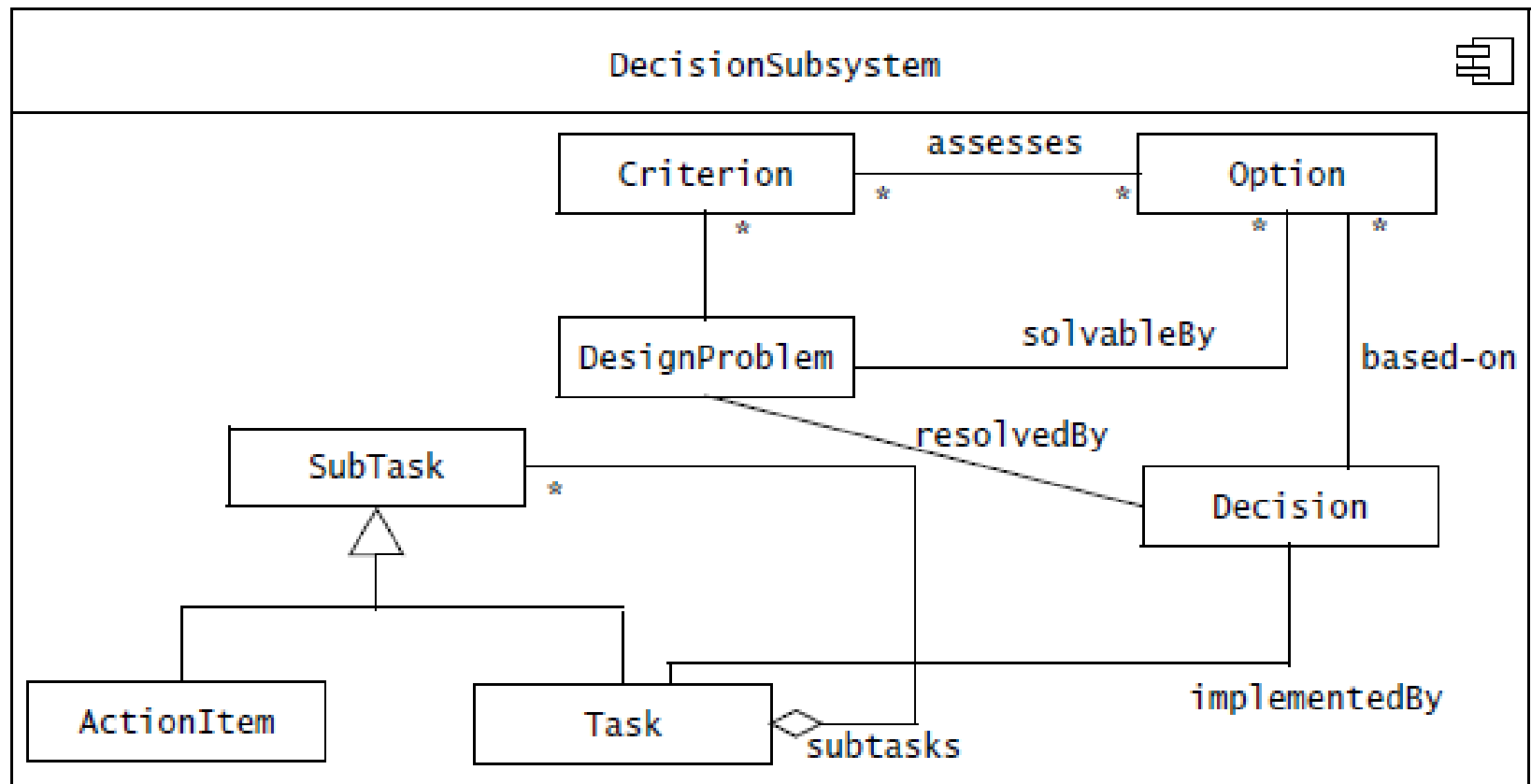
Example



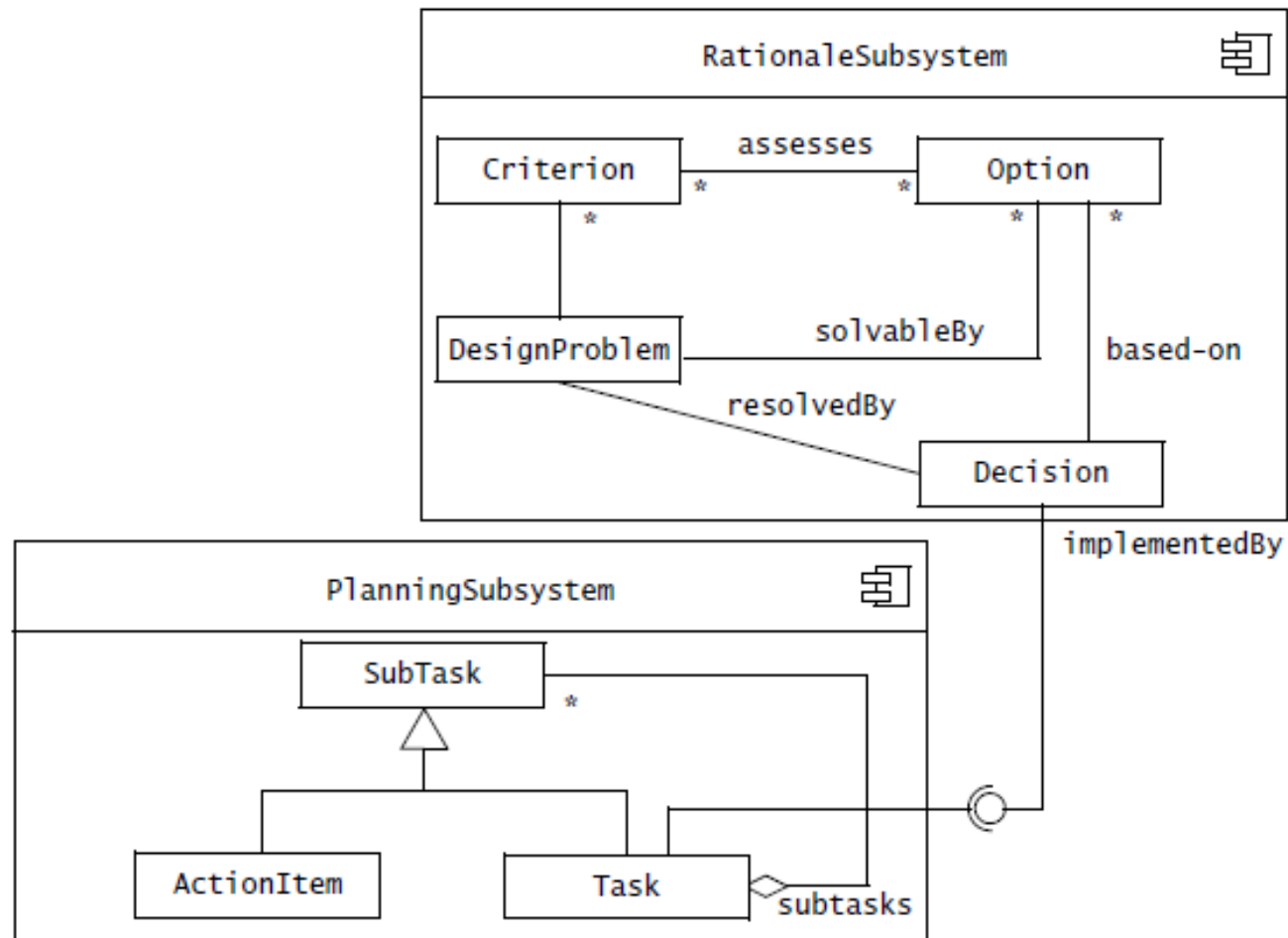
Example



Example



Component Diagram (Internal View)



Design Methodology (Recap)

- We have an abstract description of a solution to our customer's problem, a software architectural design, a plan for decomposing the design into software units and allocating the system's functional requirements to them
- No distinct boundary between the end of the architecture-design phase and the start of the module-design phase
- No comparable design recipes for progressing from a software unit's specification to its modular design
- The process taken towards a final solution is not as important as the documentation produced

Design Methodology (Recap)

- Design decisions are periodically revisited and revised
- Refactoring
 - to simplify complicated solutions or to optimize the design

References

- SE book by Pfleeger
- SE book by Pressman
- SE book by Ian Sommerville
- OOSE Book by Bruegge